



Started on	Thursday, 6 August 2026, 10:27 AM
State	Finished
Completed on	Thursday, 6 August 2026, 10:36 AM
Time taken	8 mins 53 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
```

```
{
```

```
    int i= 1;
```

```
    int s =1;
```

```
    while(s <= n)
```

```
    {
```

```
        i++;
```

```
        s += i;
```

```
    }
```

```
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

For example:

Input	Result
9	12

Answer: (penalty regime: 0 %)

```

1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      int i=1;
6      int s=1;
7      int count=2;
8      scanf("%d",&n);
9      while(s<=n)
10     {
11         count++;
12         i++;
13         count++;
14         s+=i;
15         count++;
16     }
17     count++;
18     printf("%d",count);
19     return 0;
20 }
```

	Input	Expected	Got	
✓	9	12	12	✓
✓	4	9	9	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Thursday, 6 August 2026, 11:26 AM
State	Finished
Completed on	Thursday, 6 August 2026, 11:33 AM
Time taken	6 mins 24 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
    if(n==1)
    {
        printf("*");
    }
    else
    {
        for(int i=1; i<=n; i++)
        {
            for(int j=1; j<=n; j++)
            {
                printf("*");
                printf("*");
                break;
            }
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer: (penalty regime: 0 %)

```
1  #include<stdio.h>
2  int main()
3  {
4      int n;
5      scanf("%d",&n);
6      int c=0;
7      c++;
8      if(n==1){
9          c++;
10     }
11     else{
12         for(int i=1;i<=n;i++){
13             c++;
14             for(int j=1;j<=n;j++){
15                 c++;
16                 c++;
17                 c++;
18                 break;
19             }
20             c++;
21         }
22         c++;
23     }
24     printf("%d",c);
25 }
26
```

	Input	Expected	Got	
✓	2	12	12	✓
✓	1000	5002	5002	✓
✓	143	717	717	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 24 August 2026, 7:02 PM
State	Finished
Completed on	Monday, 24 August 2026, 7:20 PM
Time taken	17 mins 18 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
Factor(num) {
{
    for (i = 1; i <= num;++i)
    {
        if (num % i== 0)
        {
            printf("%d ", i);
        }
    }
}
```

Note: No need of counter increment for declarations and scanf() and counter variable printf() statement.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1 | #include<stdio.h>
2 | int main(){
3 |     int n;
4 |     scanf("%d",&n);
5 |     int count=0;
6 |     for(int i=1;i<=n;i++){
7 |         count++;
8 |         if(n%i==0){
9 |             count++;
10 |        }
11 |        count++;
12 |    }
13 |    count++;
14 |
15 |    printf("%d ",count);
16 | }
```

	Input	Expected	Got	
✓	12	31	31	✓
✓	25	54	54	✓
✓	4	12	12	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 24 August 2026, 7:20 PM
State	Finished
Completed on	Monday, 24 August 2026, 7:24 PM
Time taken	4 mins 34 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1

Correct

Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time

complexity using counter method.

```
void function(int n)
{
    int c= 0;
    for(int i=n/2; i<n; i++)
        for(int j=1; j<n; j = 2 * j)
            for(int k=1; k<n; k = k * 2)
                c++;
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:
A positive Integer n

Output:
Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int count=1;
6      for(int i=n/2;i<n;i++){
7          count++;
8          for(int j=1;j<n;j=2*j){
9              count++;
10             for(int k=1;k<n;k=k*2){
11                 count++;
12                 count++;
13             }
14             count++;
15         }
16         count++;
17     }
18     count++;
19     printf("%d",count);
20 }
```

	Input	Expected	Got	
✓	4	30	30	✓
✓	10	212	212	✓

Passed all tests! ✓

Correct

Marks for this submission: 1.00/1.00.

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Started on	Monday, 24 August 2026, 7:24 PM
State	Finished
Completed on	Monday, 24 August 2026, 7:29 PM
Time taken	4 mins 59 secs
Marks	1.00/1.00
Grade	10.00 out of 10.00 (100%)

Question 1 | Correct Mark 1.00 out of 1.00

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
    int rev = 0, remainder;
    while (n != 0)
    {
        remainder = n % 10;
        rev = rev * 10 + remainder;
        n /= 10;
    }
    print(rev);
}
```

Note: No need of counter increment for declarations and scanf() and count variable printf() statements.

Input:

A positive Integer n

Output:

Print the value of the counter variable

Answer:

```
1  #include<stdio.h>
2  int main(){
3      int n;
4      scanf("%d",&n);
5      int count=1;
6      int rev=0,remainder;
7      while(n!=0){
8          count++;
9          remainder=n%10;
10         count++;
11         rev=rev*10+remainder;
12         count++;
13         n/=10;
14         count++;
15     }
16     count++;
17     count++;
18     printf("%d",count);
19 }
```

	Input	Expected	Got	
✓	12	11	11	✓
✓	1234	19	19	✓

Passed all tests! 

Correct

Marks for this submission: 1.00/1.00.

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